Problem 1: Finding Complexity using Counter Method

Back

Question 1 | Correct Marked out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;

    int s =1;

    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);
    int count=0;

    int i = 1;
    count++;
    int s = 1;
    count++;


    while(s <= n)
    {
        i++;
        count++;

        s += i;
        count++;

        count++;
    }
    count++;

    printf("%d", count);

    return 0;
}
```

Check

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Problem 2: Finding Complexity using Counter method

Back

Question 1 | Not complete Marked out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

```c

int main()
{
    int n;
    scanf("%d", &n);

    int count = 1;

    if(n == 1)
    {
        count++;
    }
    else
    {
        for(int i = 1; i <= n; i++)
        {
            count++;
            count++;

            for(int j = 1; j <= n; j++)
            {
                count++;
                count++;
                break;
            }

            count++;
        }

        count++;
    }

    printf("%d", count);

    return 0;
}
```

Check

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Problem 3: Finding Complexity using Counter Method

Back

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using counter method.

```
 Factor(num) {
 {
    for (i = 1; i <= num;++i)
    {
     if (num % i== 0)
        {
          printf("%d ", i);
        }
     }
  }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

er:

```c
#include <stdio.h>

int main()
{
    int num, count = 0;

    scanf("%d", &num);

    for (int i = 1; i <= num; ++i)
    {
        count++;

        if (num % i == 0)
        {
            count++;
        }

        count++;
    }
    count++;

    printf("%d", count);

    return 0;
}
```

Check

Input	Expected	Got	
12	31	31	✔
25	54	54	✔
4	12	12	✔

ed all tests! ✔

ct

for this submission: 1.00/1.00.

Problem 4: Finding Complexity using Counter Method

Back

Question 1 | Correct Marked out of 1.00 Flag question

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1
2 #include <stdio.h>
```

Answer:

```c

#include <stdio.h>

int main()
{
    int n, c = 1;
    scanf("%d", &n);

    for (int i = n / 2; i < n; i++)
    {
        c++;

        for (int j = 1; j < n; j = 2 * j)
        {
            c++;

            for (int k = 1; k < n; k = k * 2)
            {
                c++;
                c++;
            }

            c++;
        }

        c++;
    }

    c++;
    printf("%d", c);
    return 0;
}
```

Check

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Problem 5: Finding Complexity using counter method

Back

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c

#include <stdio.h>

int main()
{
    int n, rev = 0, remainder, count = 0;
    scanf("%d", &n);

    count++;
    count++;

    while (count++, n != 0)
    {
        remainder = n % 10;
        count++;

        rev = rev * 10 + remainder;
        count++;

        n /= 10;
        count++;
    }

    printf("%d", count);
    return 0;
}
```

Check

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.